Meninges and Ventricular System

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Learning Objectives (2026)

1. Describe the anatomy and organization of the meninges, ventricular system, and cerebrospinal fluid pathways, and relate their structural relationships to normal central nervous system protection and function. (CLO 1)
2. Interpret the meninges, ventricular system, and cerebrospinal fluid spaces using cross-sectional anatomy and medical imaging, and correlate normal imaging anatomy with common neurological disorders. (CLO 1, CLO 5)
3. Explain the pathophysiology, clinical features, and diagnostic principles of hydrocephalus and other disorders involving cerebrospinal fluid circulation and the meninges. (CLO 3)

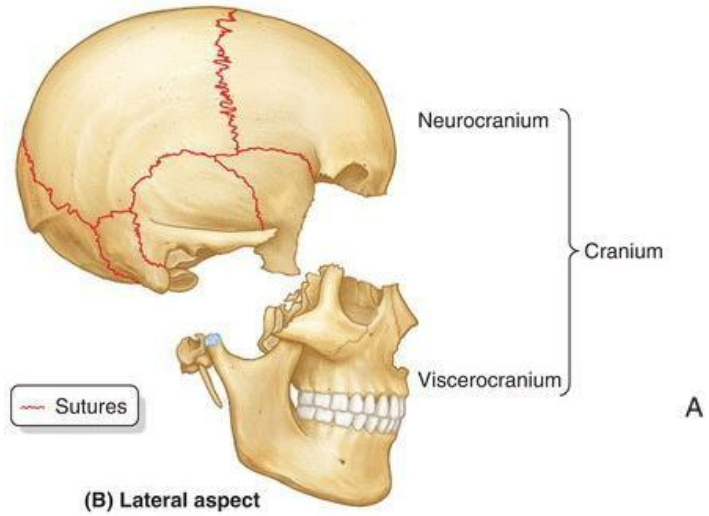
Pre-Session Assignment

- Preview the PowerPoint slides. You are not expected to memorize details before class.
- Watch:
  - B&B: Ventricles and Sinuses (22 min)
  - B&B: Blood Brain Barrier (12 min)

Skull

Head Osteology

- 22 bones skull bones: 8 neurocranial + 14 facial
- +6 auditory ossicles
- Neurocranium: calvaria, cranial base
- Viscerocranium: facial skeleton



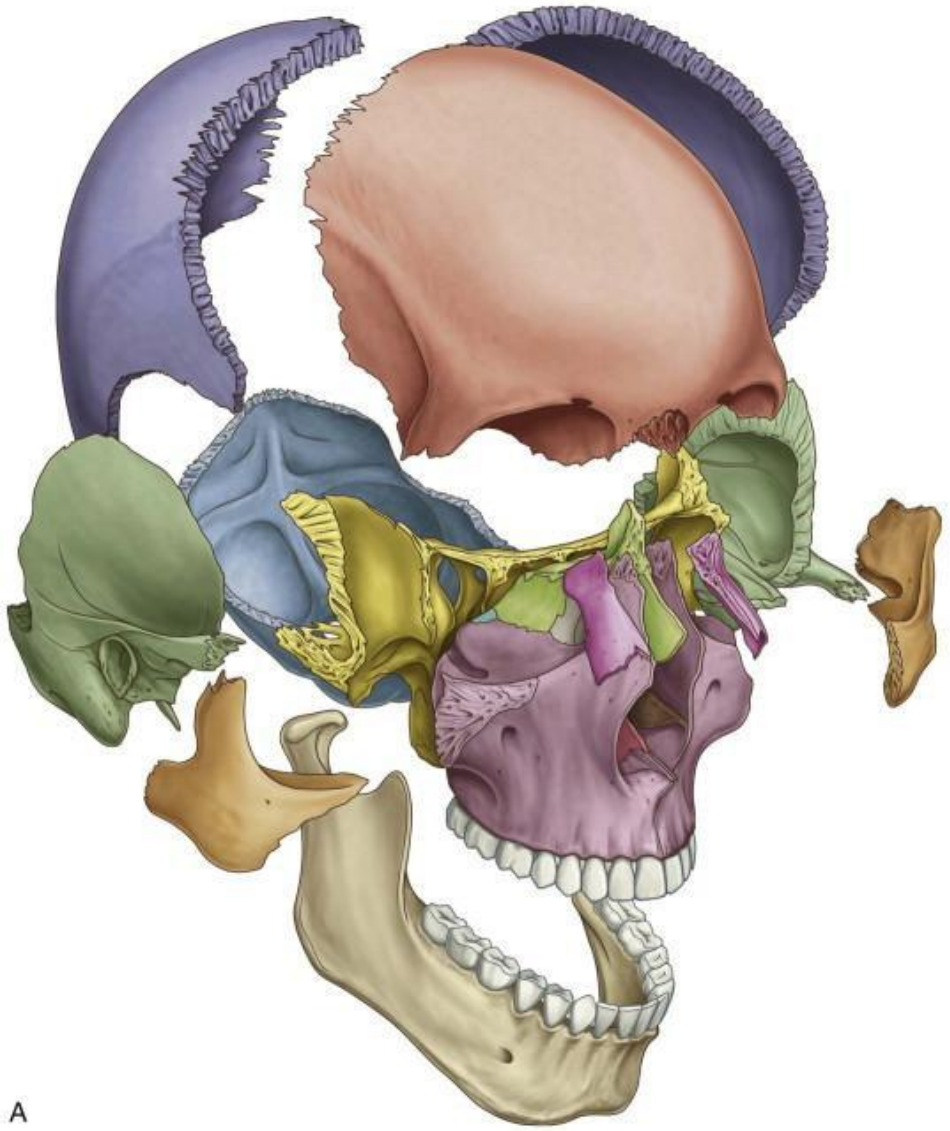
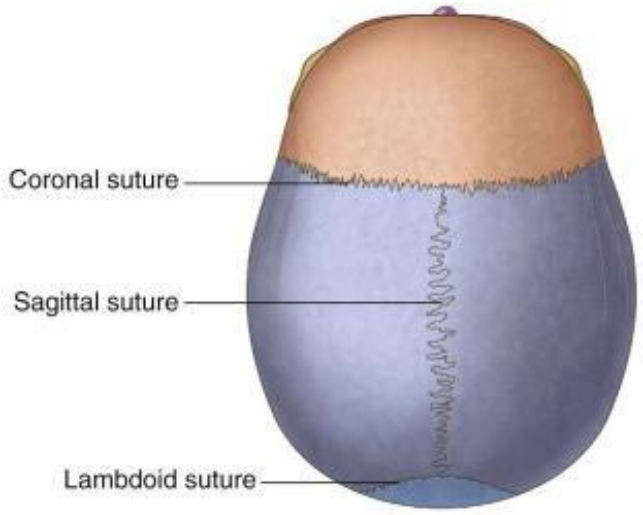
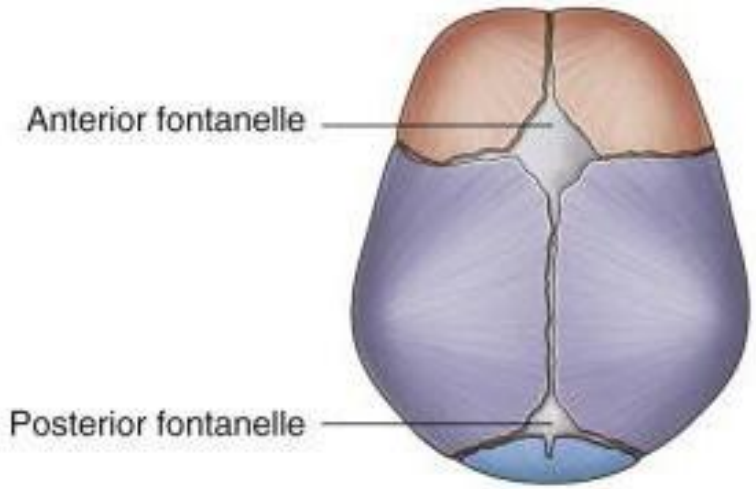


Fig. 8.1B Adult Cranium (lateral). COA, 8e.
Fig. 8.7 A. Skull: Bones. GAS, 4e.

Skull Sutures

- Immobile joints: coronal, sagittal, lambdoid
- Junctions: bregma, lambda, pterion, asterion
- Clinical: fontanelles (newborn), metopic suture





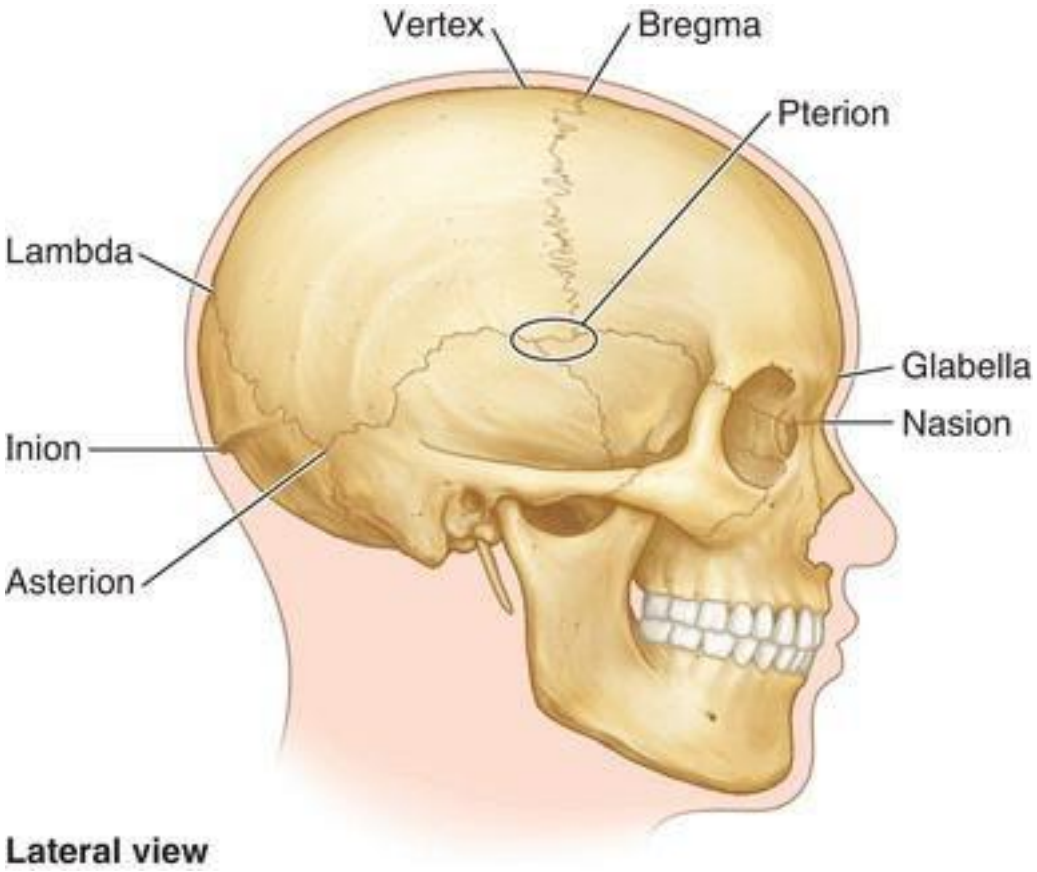
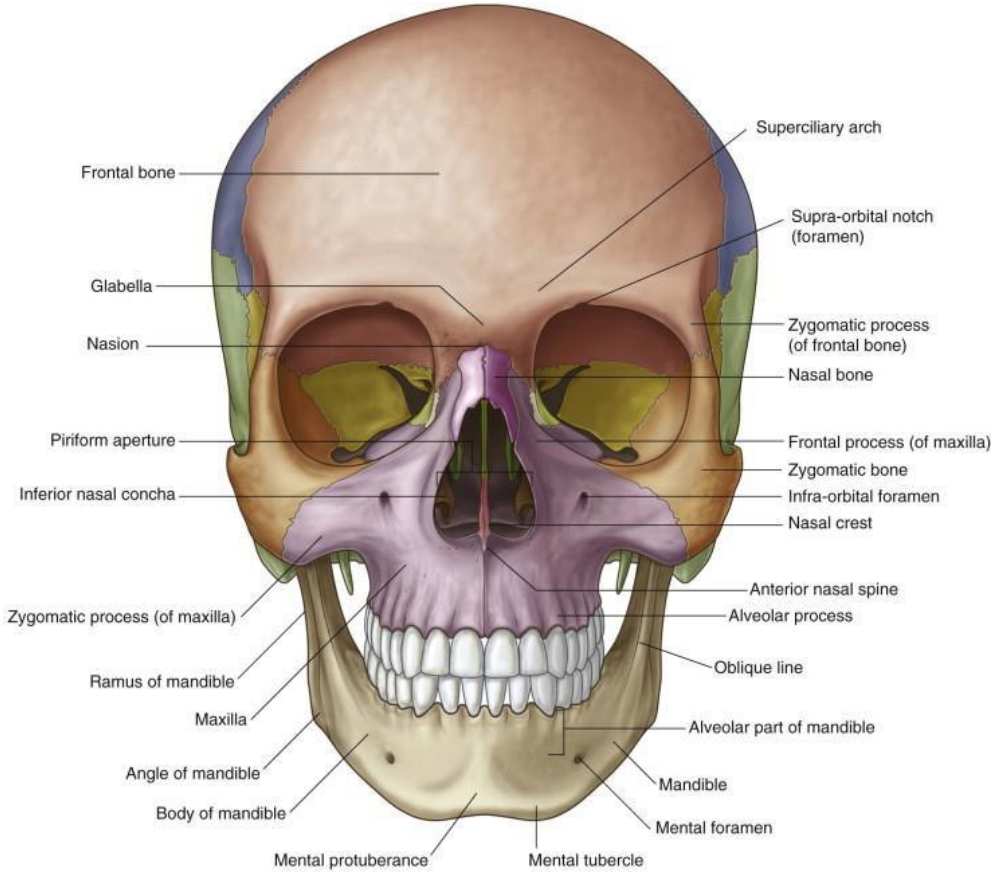


Fig. 8.7C. Fontanelles and lambdoid suture. GAS, 4e.
Fig. 8.6 Craniometric points. COA, 8e.

Skull Anterior and Posterior Views

- Frontal, nasal, zygomatic, maxilla, mandible anteriorly
- Occipital and parietal bones posteriorly, lambdoid suture



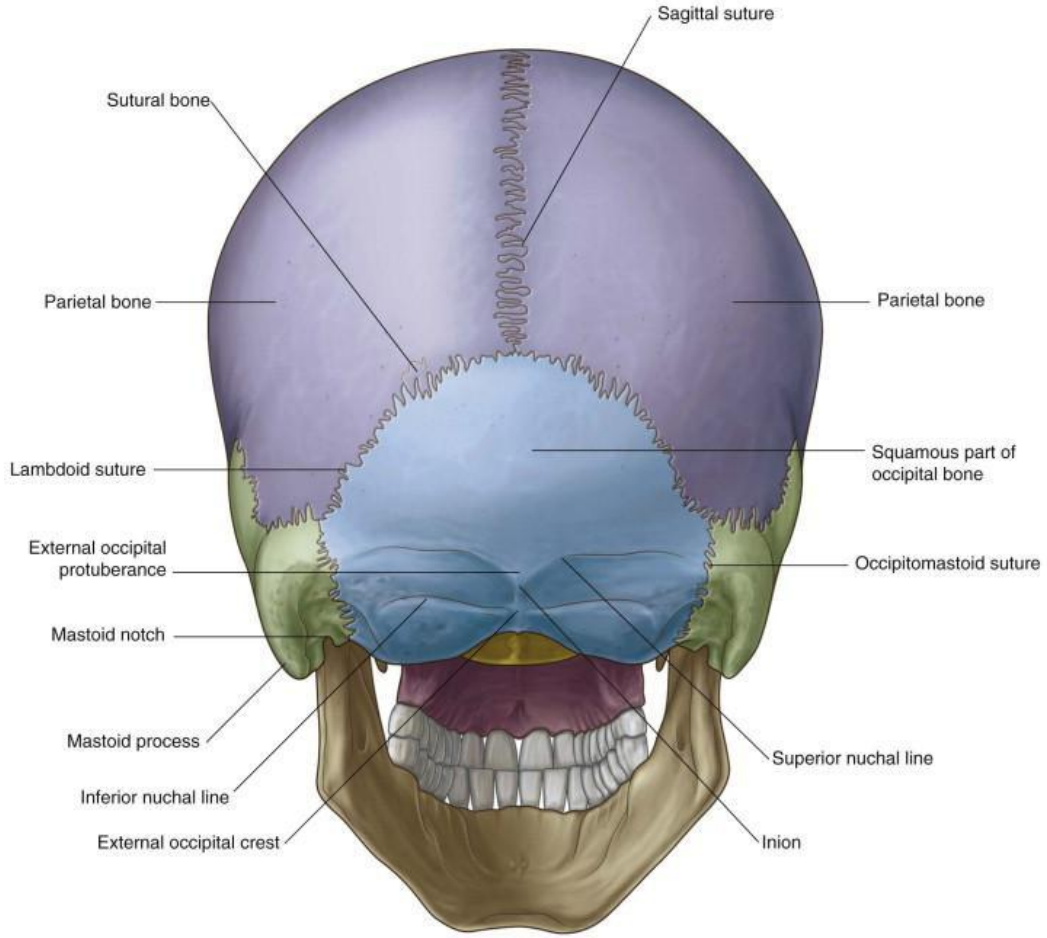
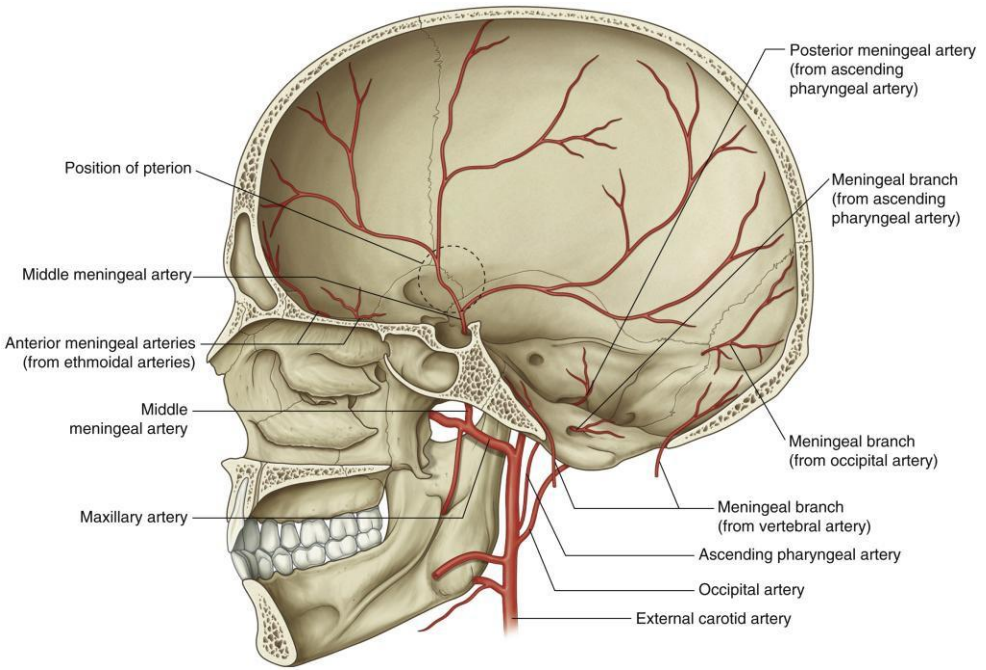


Fig. 8.18 Anterior view of the skull.
Fig. 8.20 Posterior view of the skull
Gray's Anatomy for Students, 4e.

Skull Lateral View

- Temporal and parietal bones
- Greater wing of sphenoid
- Zygomatic arch
- Pterion: middle meningeal artery lies deep to it



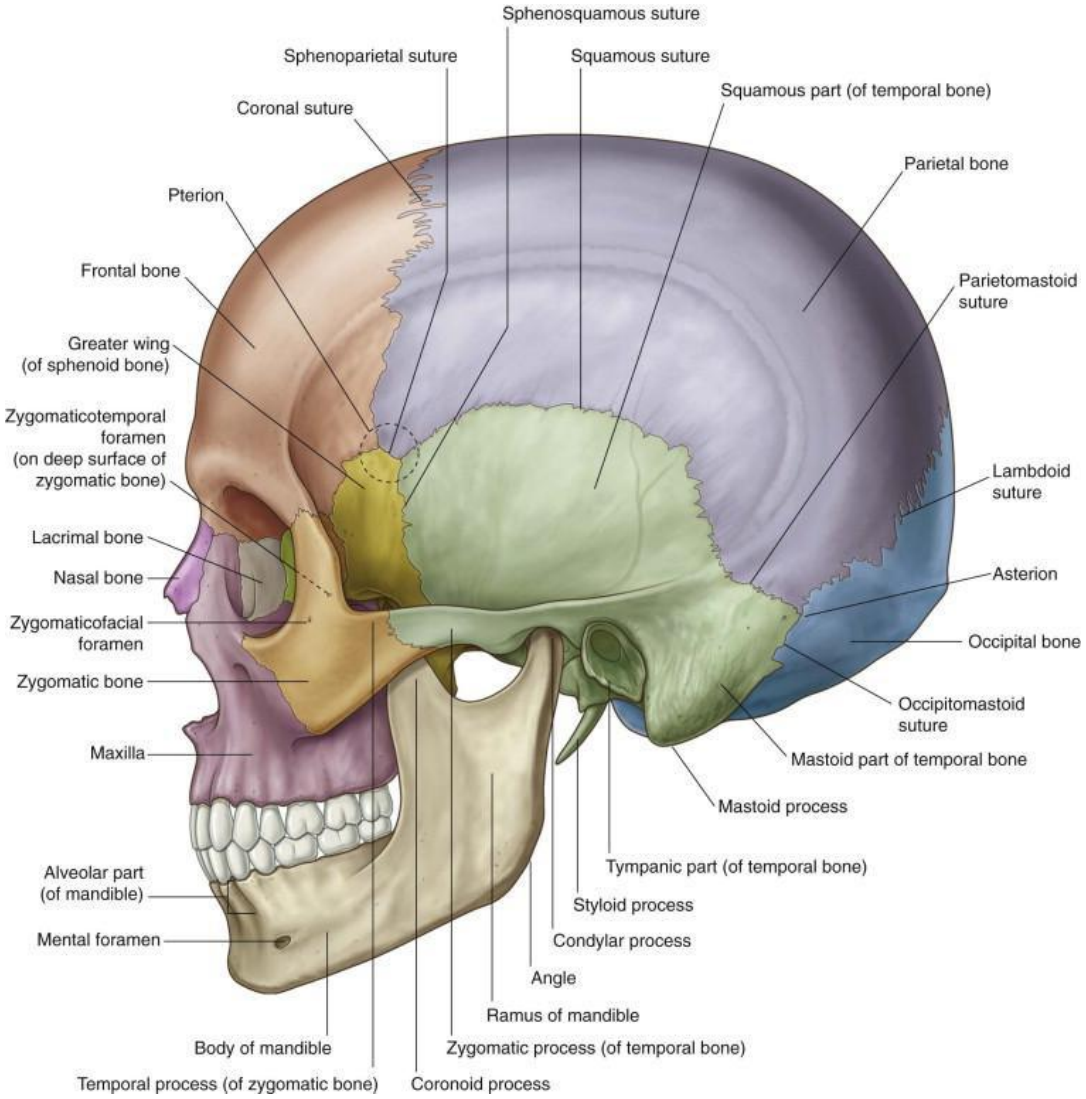


Fig. 8.19 Lateral view of the skull
Gray's Anatomy for Students, 4e.

Skull Superior View

- Calvaria: Frontal, parietal bones, sagittal suture

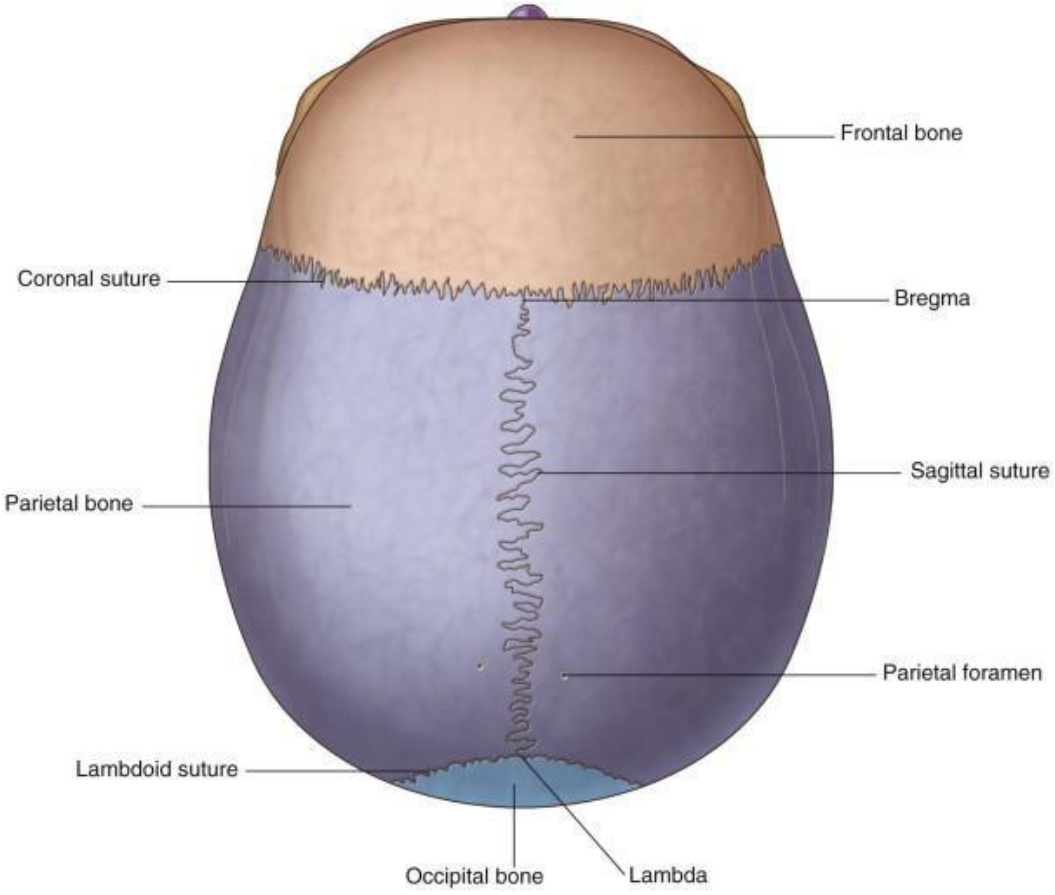


Fig. 8.21 Superior view of the skull. GAS, 4e.

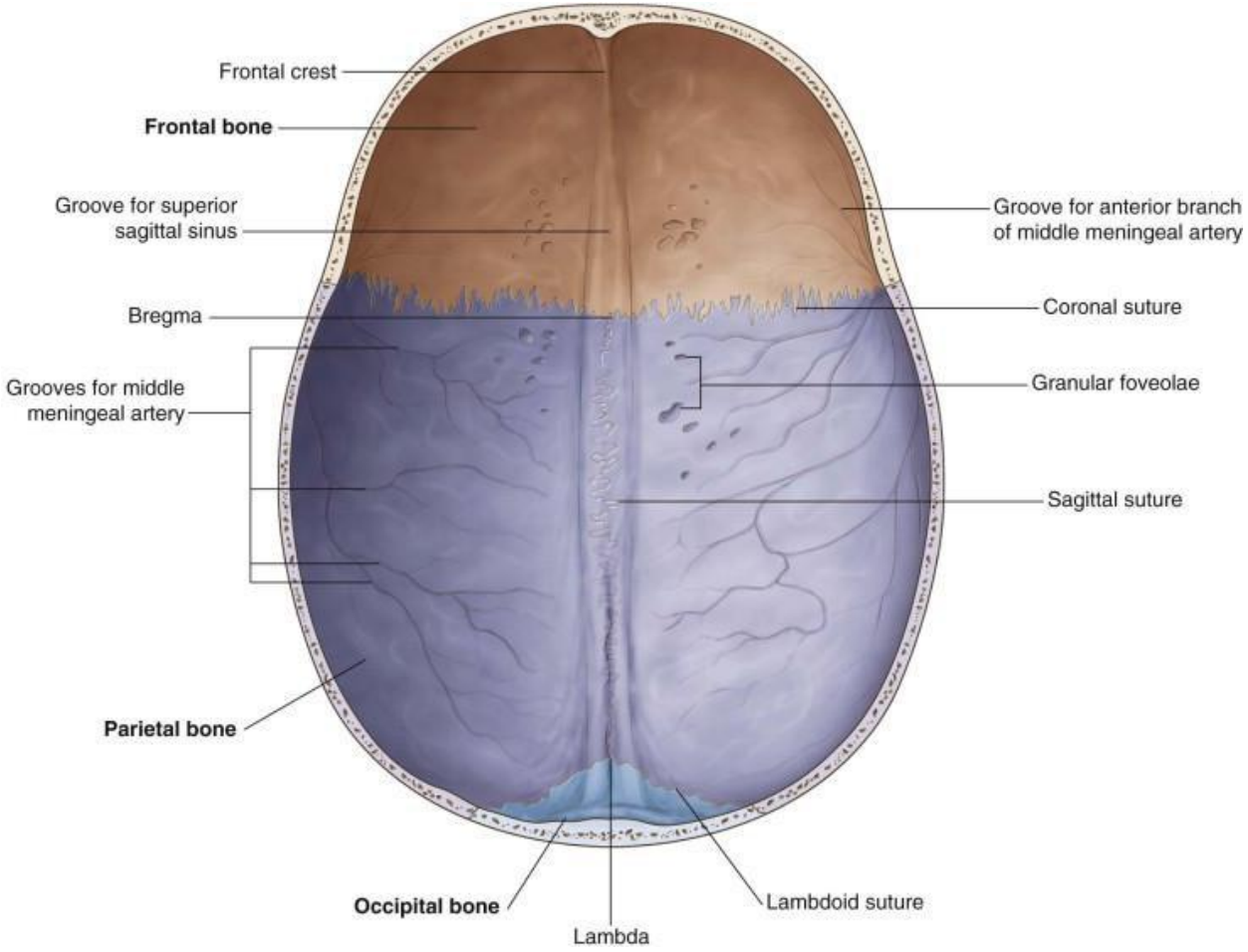
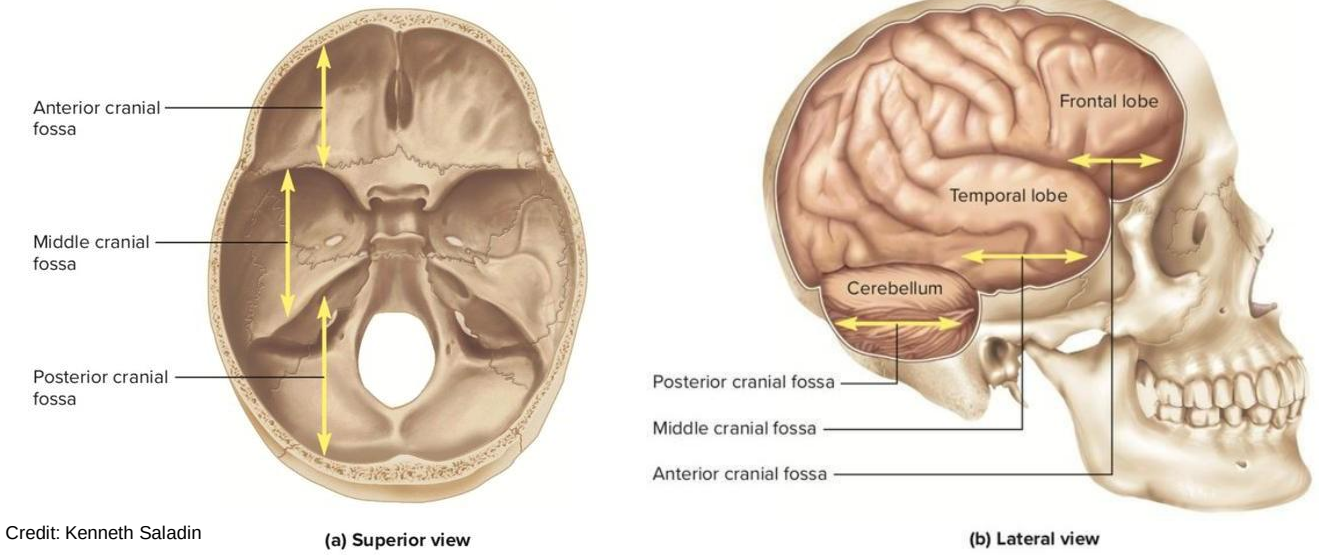


Fig. 8.24 Roof of the cranial cavity. GAS, 4e.

Cranial Floor (Fossae Overview)

- Anterior: frontal lobes
- Middle: temporal lobes
- Posterior: cerebellum, brainstem



Credit: Kenneth Saladin

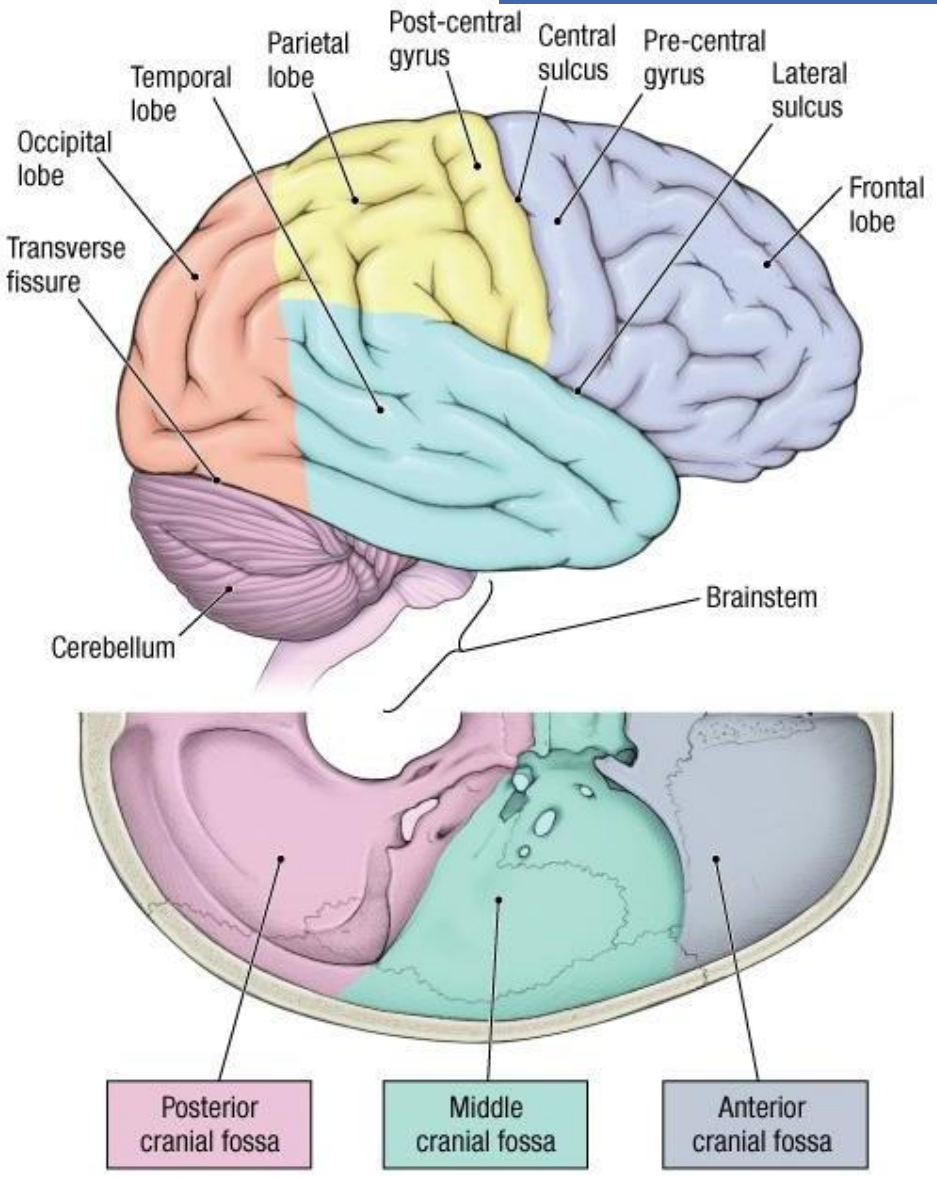


Fig. 7.41 Brain and its relationship to the three cranial fossae
Grant's Dissector, 16e.

Cranial Floor

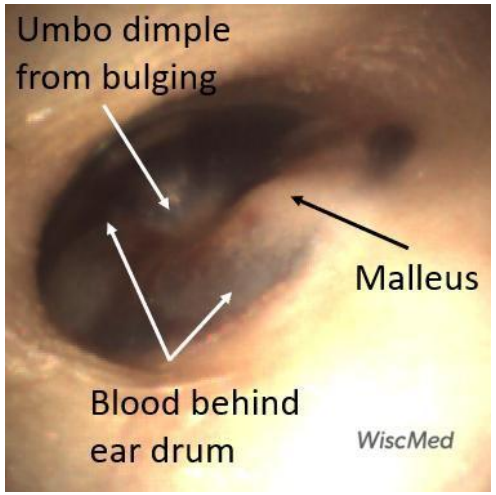
Basilar Skull Fracture

- Fractures involving the skull base
- Commonly involves the temporal bone

Clinical Signs

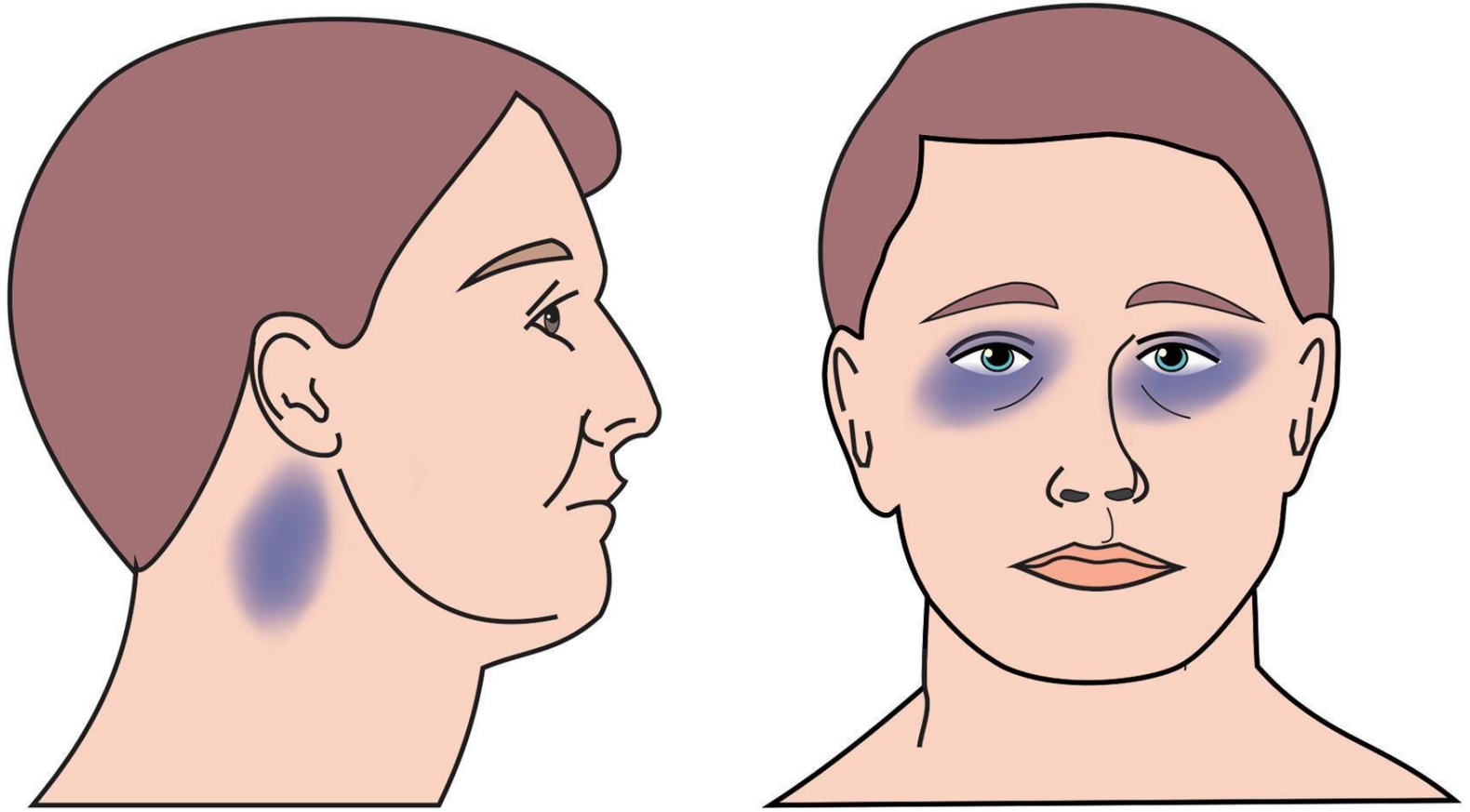
- Battle sign: postauricular/mastoid ecchymosis
- Hemotympanum: blood behind the tympanic membrane
- Raccoon eyes: periorbital ecchymosis
- CSF leakage: CSF rhinorrhea or otorrhea







Battle Sign and Raccoon Eyes



Battle sign ➔ basilar skull fracture ➔ postauricular/mastoid ecchymosis

Raccoon sign ➔ anterior cranial fossa fracture ➔ periorbital ecchymosis

Image: Piedade. Facial Trauma.
In: The Sports Medicine Physician. Springer 2019.

Meninges

Meninges

- Dura mater – outermost
- Arachnoid mater – middle
- Pia mater – innermost

Leptomeninges = arachnoid + pia

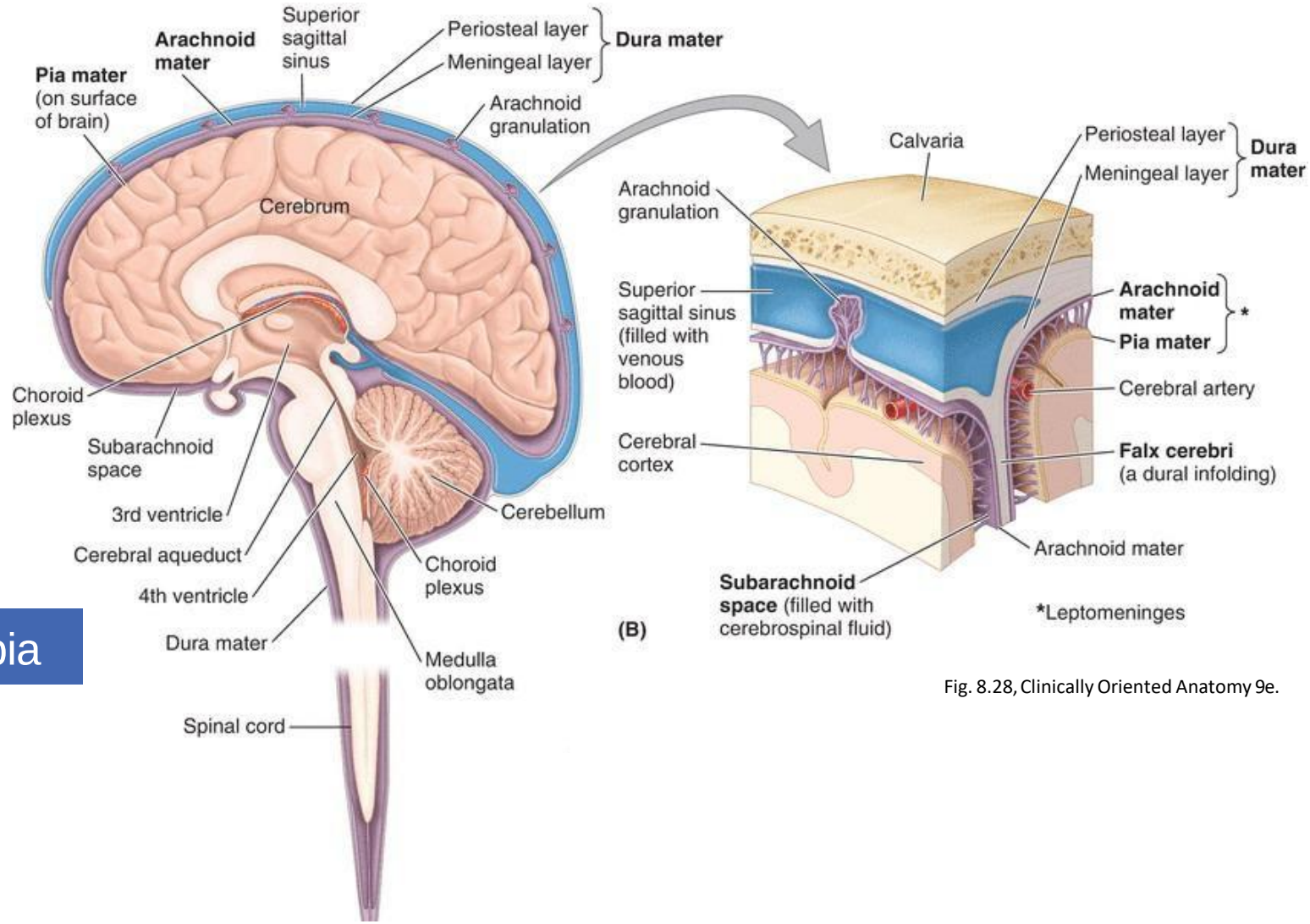


Fig. 8.28, Clinically Oriented Anatomy 9e.

Dura Mater

- Thick, tough outer meningeal layer
- **Cranial dura:** periosteal (outer) + meningeal (inner) layers
- **Spinal dura:** meningeal layer only
- Meningeal layer forms dural partitions

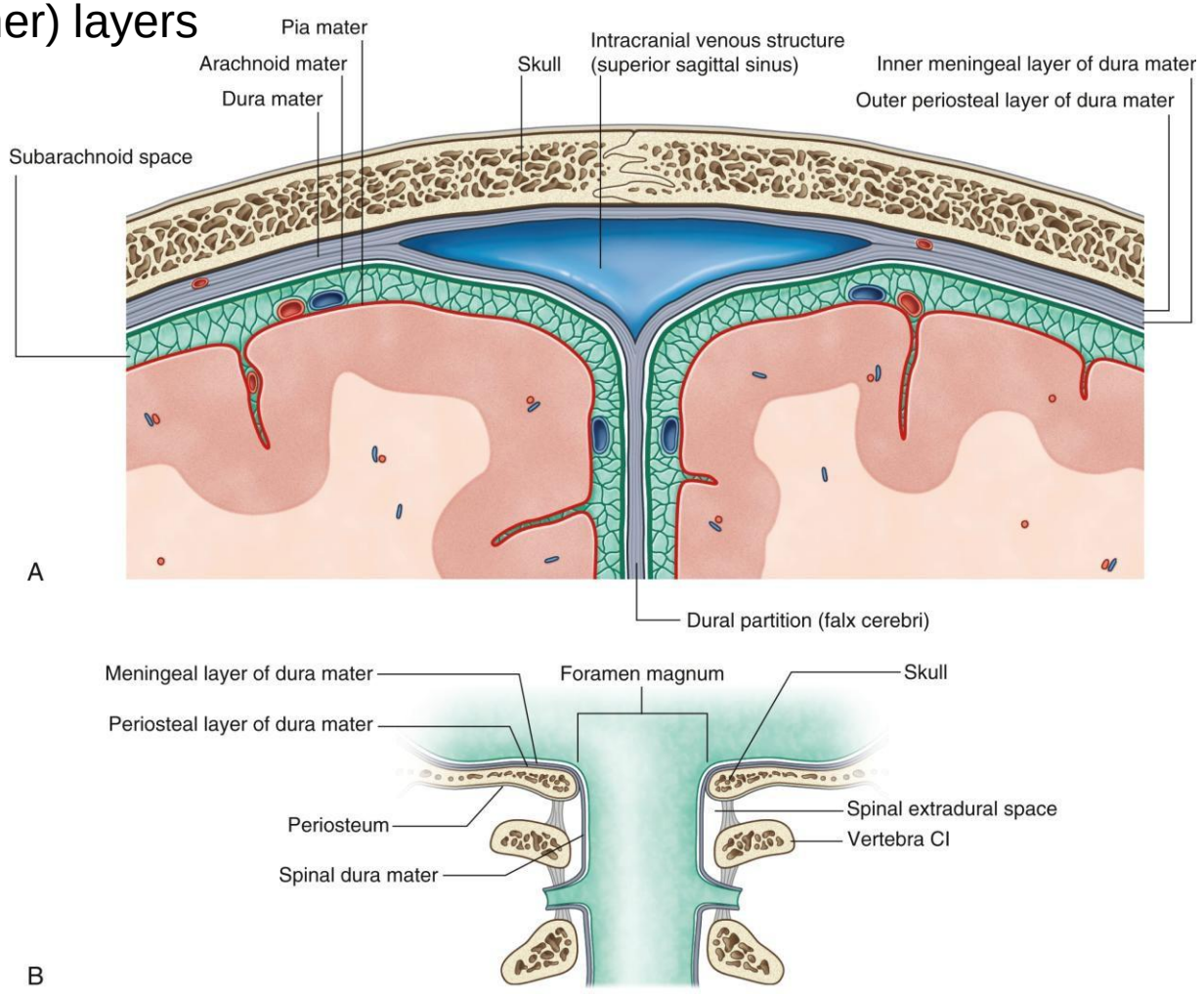


Fig. 8.31A Dural partitions. GAS, 5e.

Dural Partitions

- Dural folds that subdivide the cranial cavity
- **Falx cerebri:** separates cerebral hemispheres
- **Tentorium cerebelli:** separates cerebrum from cerebellum
- **Falx cerebelli:** separates cerebellar hemispheres
- **Diaphragma sellae:** covers pituitary gland in sella turcica

Supratentorial = above the tentorium cerebelli
Infratentorial = below the tentorium cerebelli

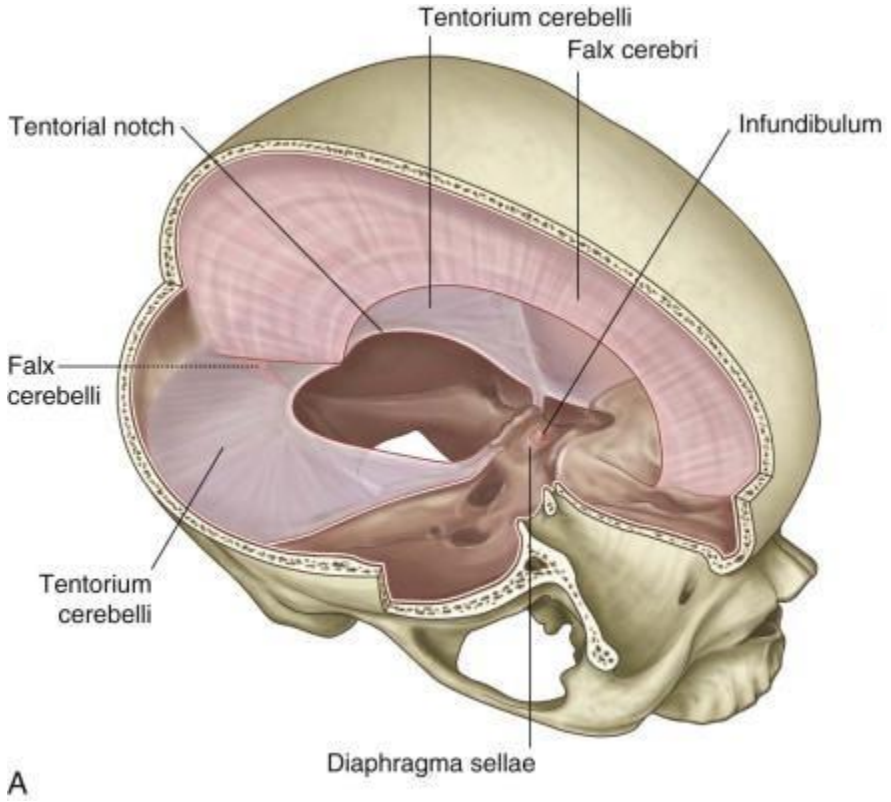


Fig. 8.32 Dural partitions.
Gray's Anatomy for Students, 4e.

Dural Innervation

Supratentorial Dura

- Primarily trigeminal nerve (CN V)
- Irritation ➔ referred pain to trigeminal distribution (forehead, face)

Posterior Fossa Dura

- Primarily upper cervical nerves (C1–3)
- Irritation ➔ referred pain to necck/occiput

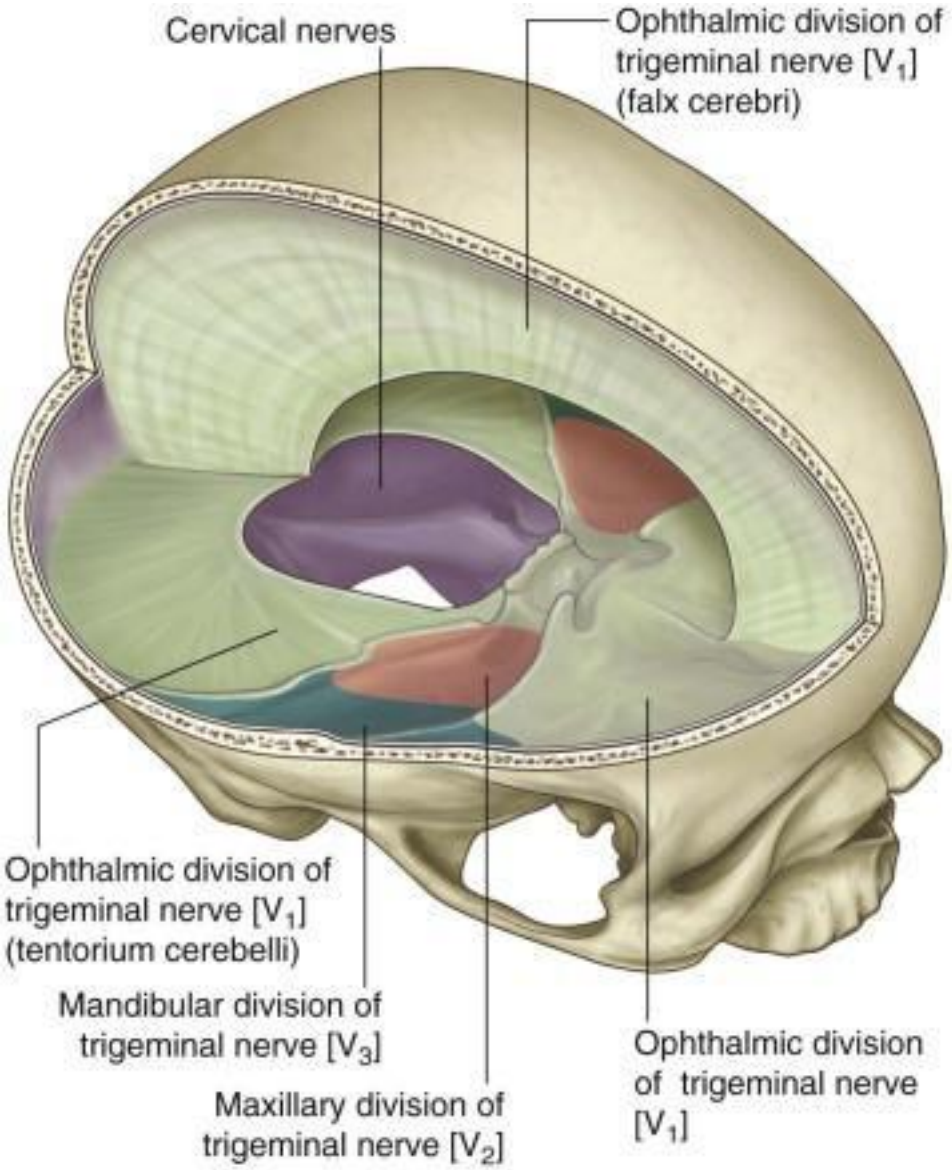


Fig. 8.34 Dural innervation
Gray's Anatomy for Students, 4e.

Arachnoid and Pia

Arachnoid Mater

- Middle meningeal layer; avascular
- Does not follow gyri and sulci
- Subarachnoid space contains CSF and cerebral vessels
- Arachnoid granulations → CSF drainage into dural venous sinuses

Pia Mater

- Innermost, delicate vascular membrane
- Adheres closely to brain and spinal cord
- Follows gyri and sulci

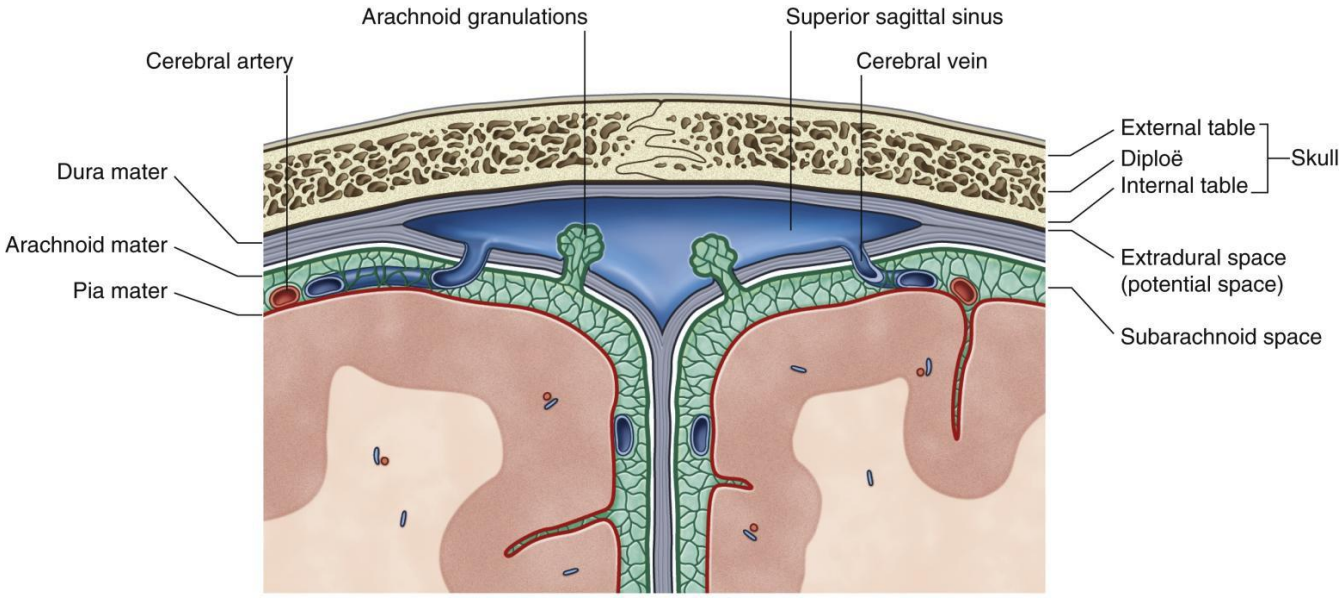
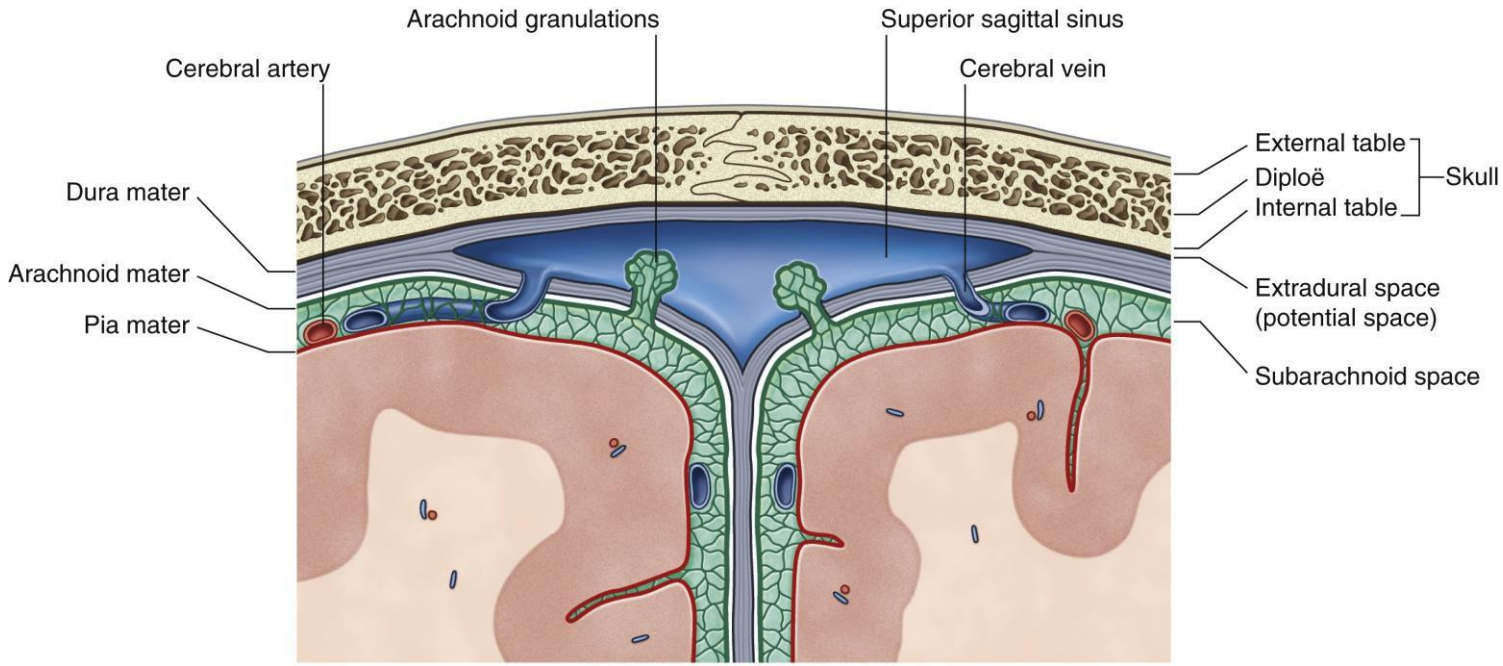


Fig. 8.35 Arrangement of meninges and spaces. GAS 5e.

Meningeal Spaces



Space	Type	Normal Contents	Clinical
Extradural (epidural)	Potential	None	MMA rupture ➔ epidural hematoma
Subdural	Potential	None	Bridging vein tear ➔ subdural hematoma
Subarachnoid	Real	CSF, vessels	Berry aneurysm rupture ➔ SAH

Fig. 8.35 Arrangement of meninges and spaces.
Gray's Anatomy for Students, 5e.

Intracranial Hematomas

Extradural (Epidural) Hematoma

- Middle meningeal artery rupture (pterion fracture)
- Lucid interval ➔ rapid deterioration
- Biconvex lens-shaped hematoma; does not cross sutures

Subdural Hematoma

- Bridging vein tear
- Risk” elderly (brain atrophy) and infants (shaken babies)
- Crescent-shaped hematoma; crosses sutures

Subarachnoid Hemorrhage

- Ruptured berry aneurysm (commonly anterior communicating artery)
- Thunderclap headache (“worst headache of my life”), meningeal signs
- CT: blood in basal cisterns/sulci

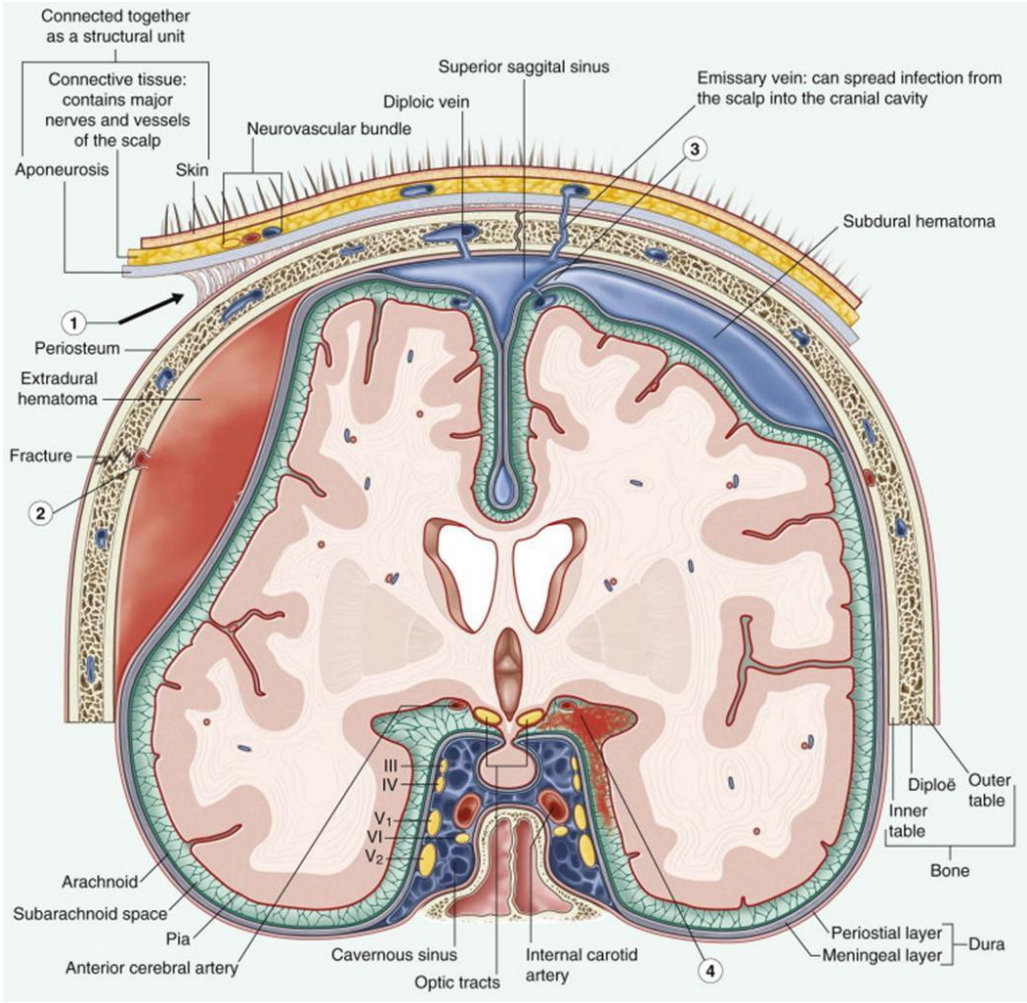


Fig. 8.49 Scalp and meninges
Gray’s Anatomy for Students, 4e.

Epidural (Extradural) Hematoma

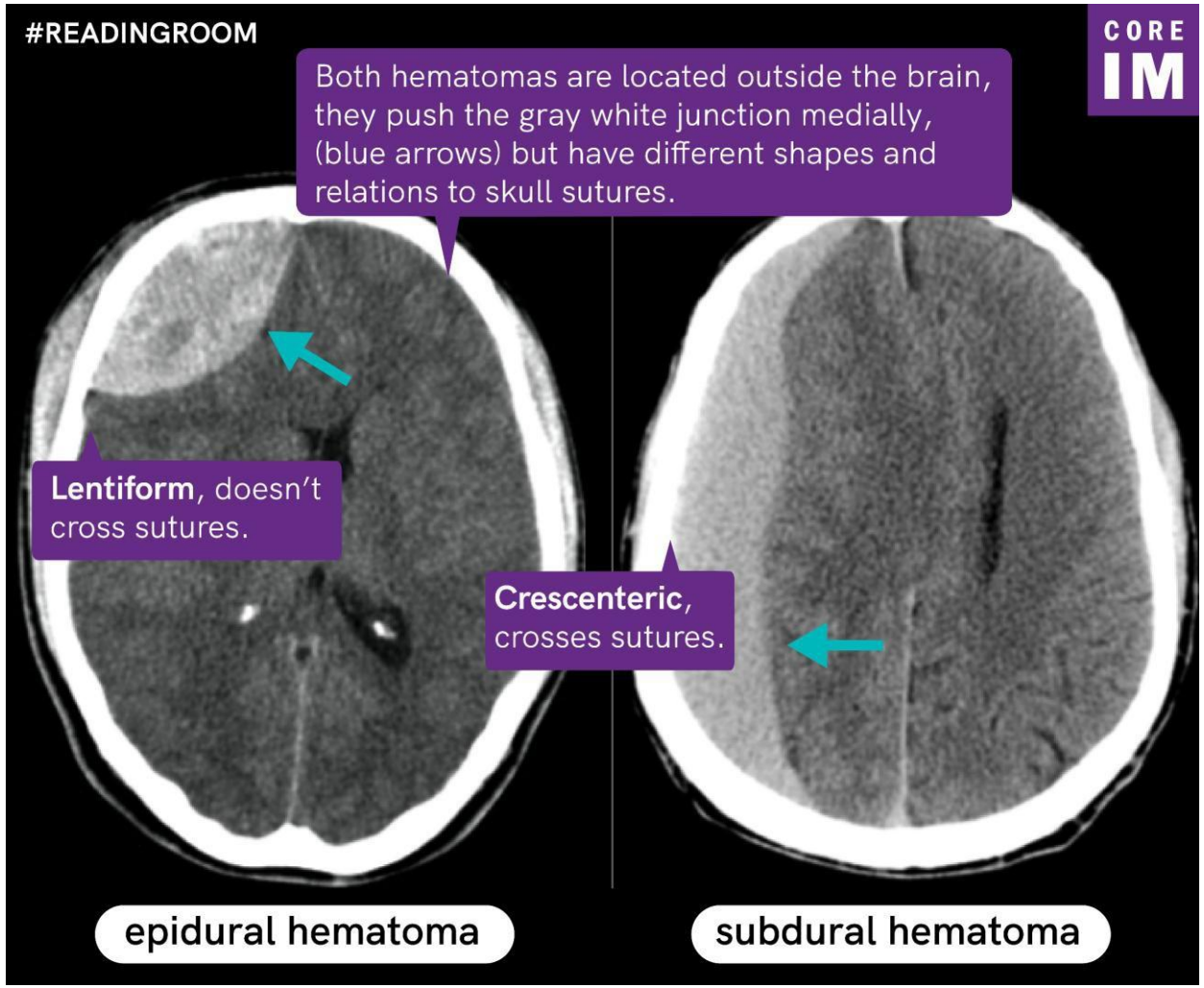
- Lens-shaped (biconvex) hematoma
- Limited by sutures (does not cross)
- Displaces brain tissue medially

Subdural Hematoma

- Crescent-shaped hematoma
- Crosses sutures, but not midline falx
- More diffuse, spreads along brain surface

Lens = Epidural

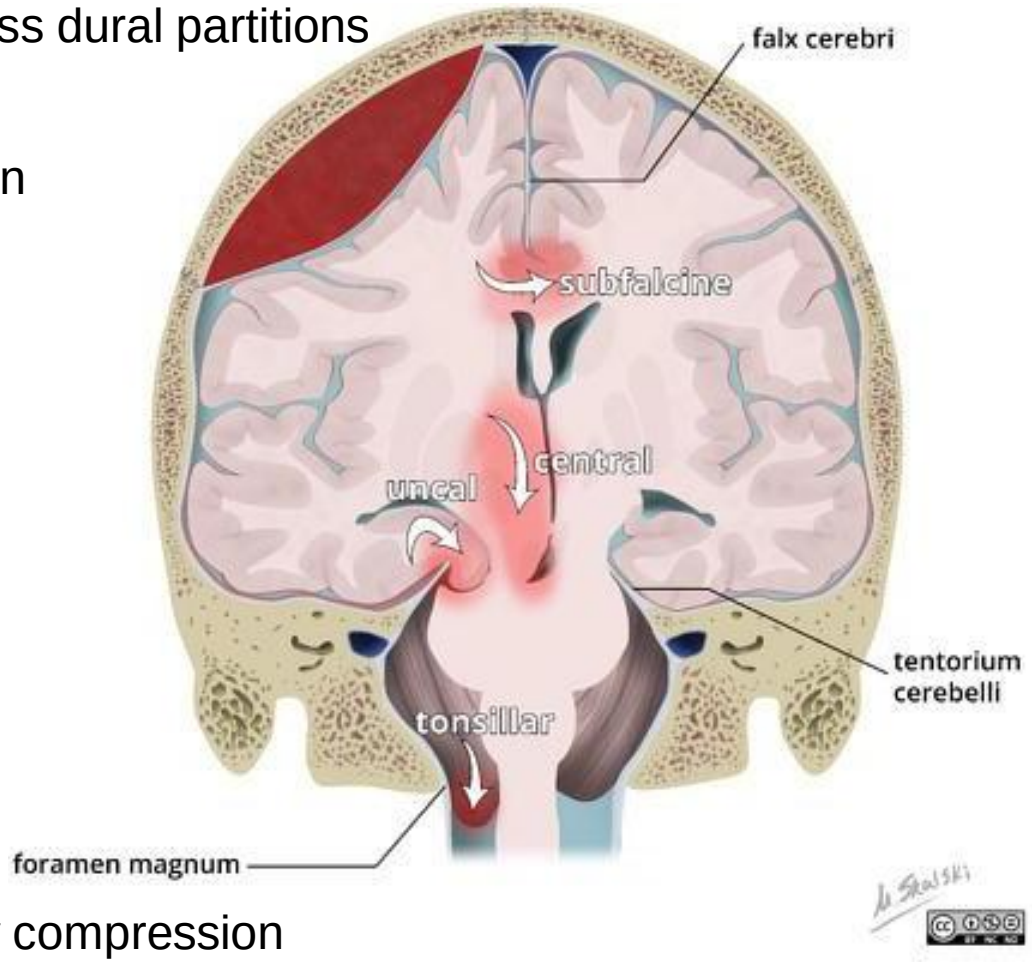
Crescent = Subdural



Herniation Syndromes

↑ Intracranial pressure ➔ displacement of brain tissue across dural partitions

Subfalcine: Cingulate gyrus under falx ➔ ACA compression



Central (transtentorial)

- Downward displacement ➔ brainstem compression

Uncal

- ipsilateral CN III compression ➔ dilated pupil;
- cerebral peduncle compression ➔ hemiparesis

Tonsillar

- cerebellar tonsils through foramen magnum ➔ medullary compression

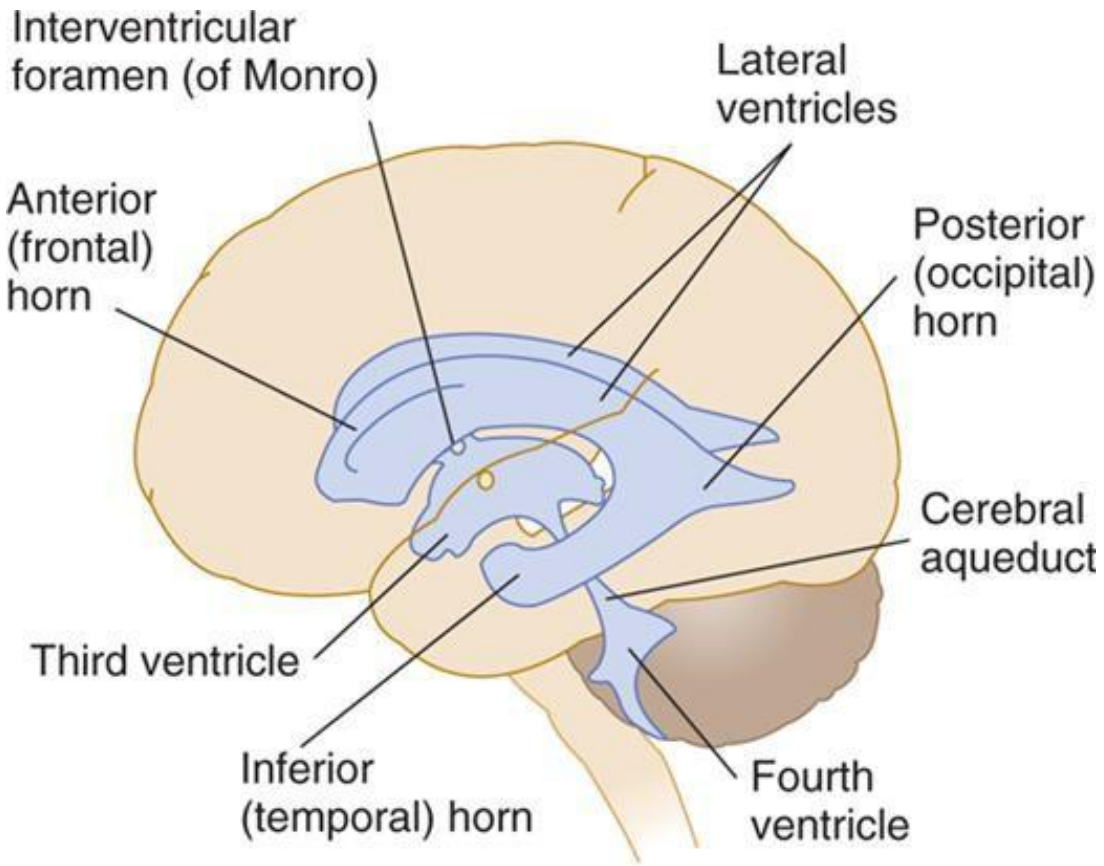
Complications:

Brainstem compression ➔ coma, cardiorespiratory failure, death

CSF & Ventricular System

Ventricular System

- Interconnected CSF-filled cavities
- Lined by ependymal cells
- Four ventricles: two lateral, third, fourth
- Fourth ventricle continues with the central canal of the spinal cord

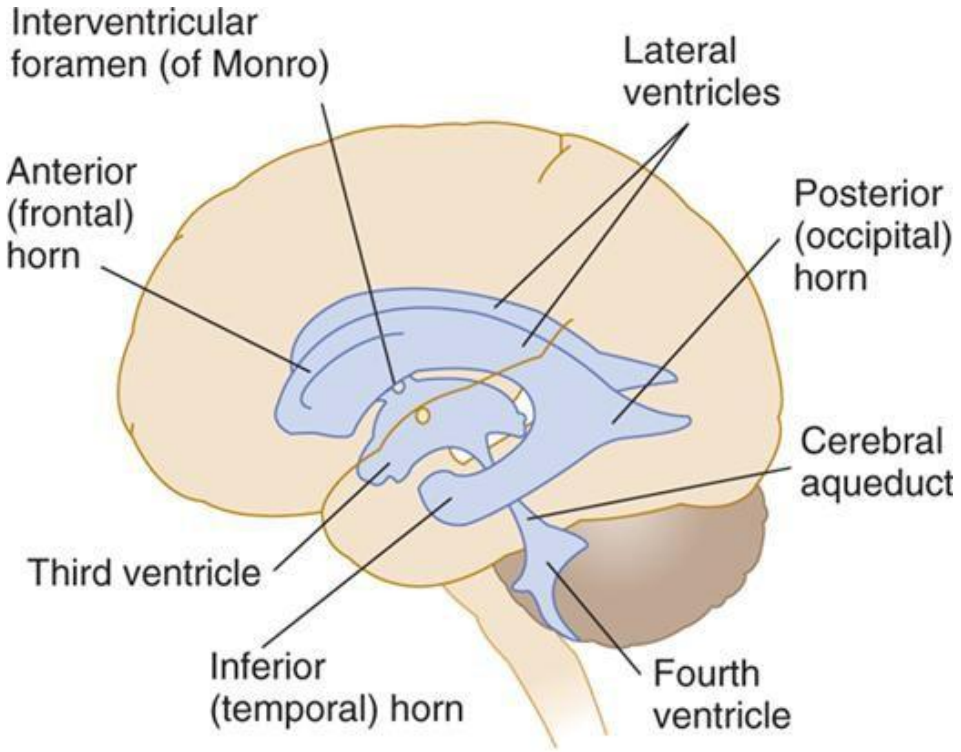


Source: Stephen G. Waxman: *Clinical Neuroanatomy, 30th Edition*
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Fig. 11-1. Ventricular System. Waxman, 30e.

Lateral Ventricles

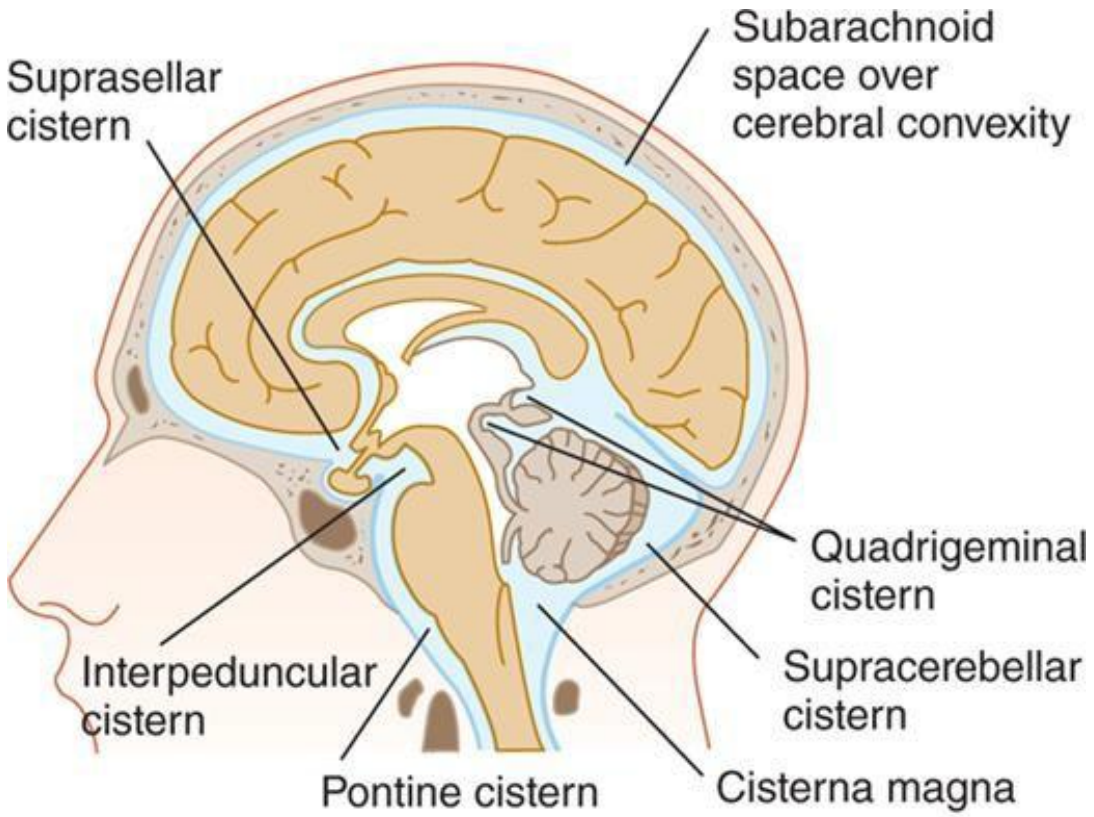
- C-shaped ventricles in cerebral hemispheres
- Consist of body + 3 horns (anterior, posterior, inferior)
- Communicate with the third ventricle via the interventricular foramina (of Monro)



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Third and Fourth Ventricles

- Third ventricle: between the thalami
- 4th ventricle: between brainstem and cerebellum
- Connected by the cerebral aqueduct
- CSF exits the 4th ventricle into the subarachnoid space through:
- Two lateral foramina (Luschka)
- One median foramen (Magendie)



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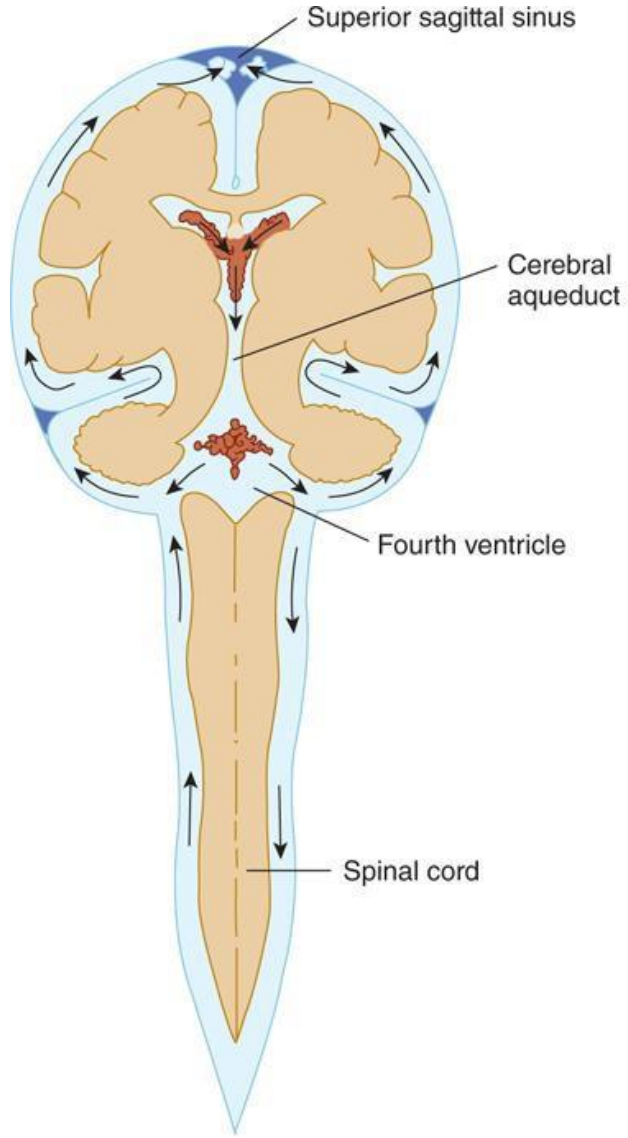
Cerebrospinal Fluid (CSF)

- Clear, colorless fluid produced primarily by choroid plexus
- Circulates through ventricles ➔ subarachnoid space
- Reabsorbed through arachnoid granulations ➔ dural venous sinuses
- drains into superior sagittal sinus via arachnoid villi
- Functions: buoyancy, protection, and CNS homeostasis

Choroid Plexus

- Vascular tissue covered by modified ependymal cells
- Present in the ventricles
- Main site of CSF production

Lateral ventricles → interventricular foramina → 3rd ventricle → cerebral aqueduct → 4th ventricle → Luschka/Magendie → subarachnoid space → arachnoid granulations → dural venous sinuses



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Hydrocephalus

Definition

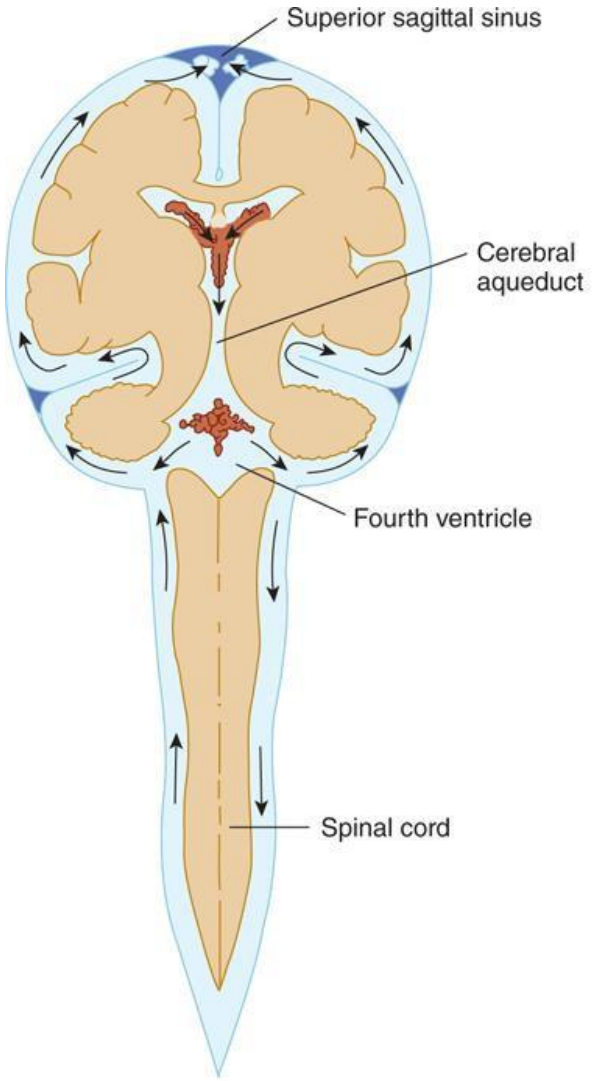
- Abnormal accumulation of CSF ➔ ventricular dilation ± increased ICP

Causes

- Obstruction of CSF flow ➔ non-communicating (obstructive)
- Impaired CSF absorption ➔ communicating
- CSF overproduction ➔ rare (e.g., choroid plexus papilloma)

Clinical Features

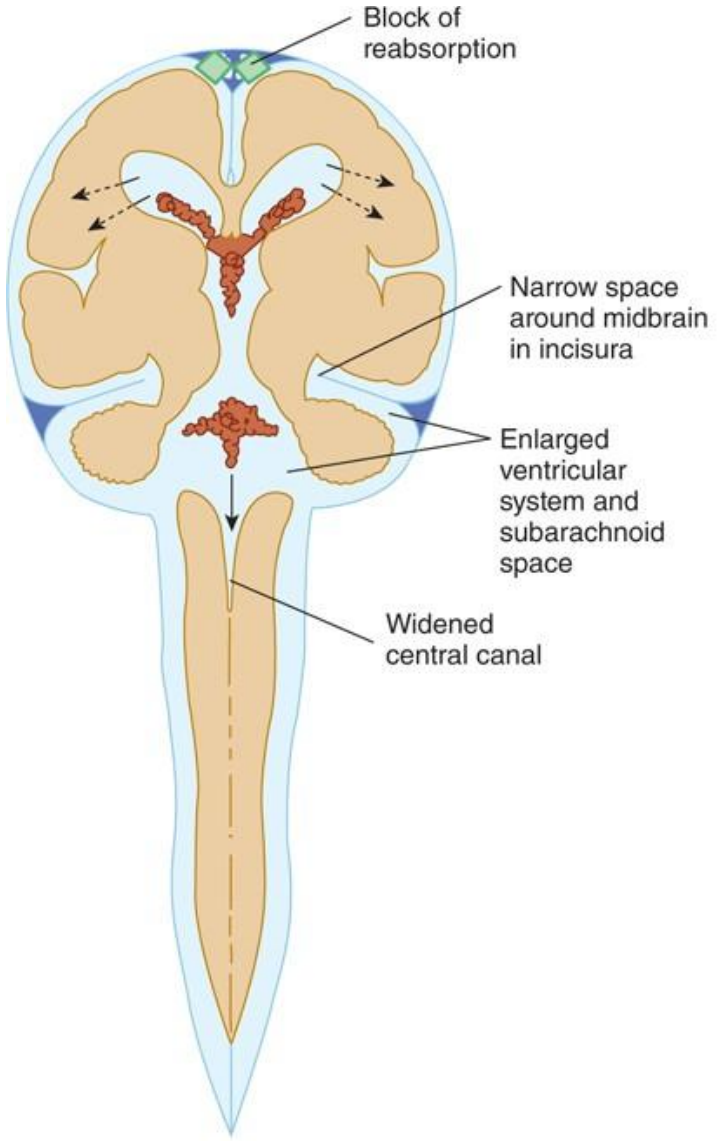
- Headache, nausea/vomiting, papilledema
- Cognitive or neurologic deficits
- Infants: increasing head circumference, bulging fontanelle



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Communicating Hydrocephalus

- No obstruction within the ventricular system
- ↓ CSF absorption at arachnoid granulations
- ***All ventricles enlarge***
- Causes: SAH, meningitis
- May cause ↑ ICP



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Non-Communicating (Obstructive) Hydrocephalus

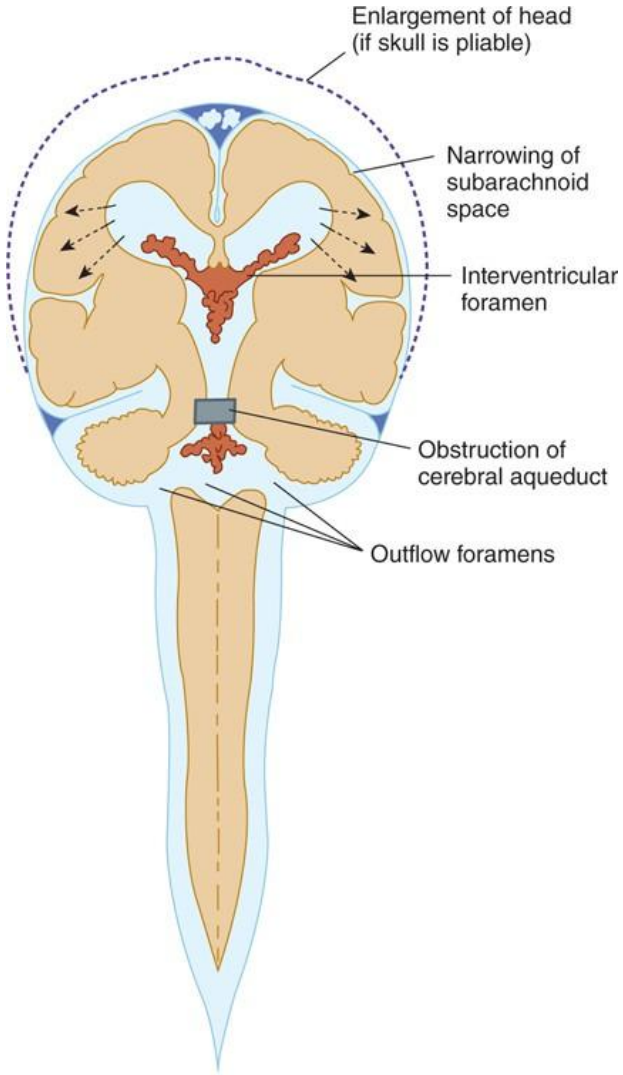
- Obstructive within the ventricular system
- ***Ventricles proximal to the obstruction enlarge***

Common Sites of Obstruction

- Interventricular foramen
- Cerebral aqueduct (most common)
- Fourth ventricular outlets

Causes

- Congenital malformations, tumors, cysts, hemorrhage



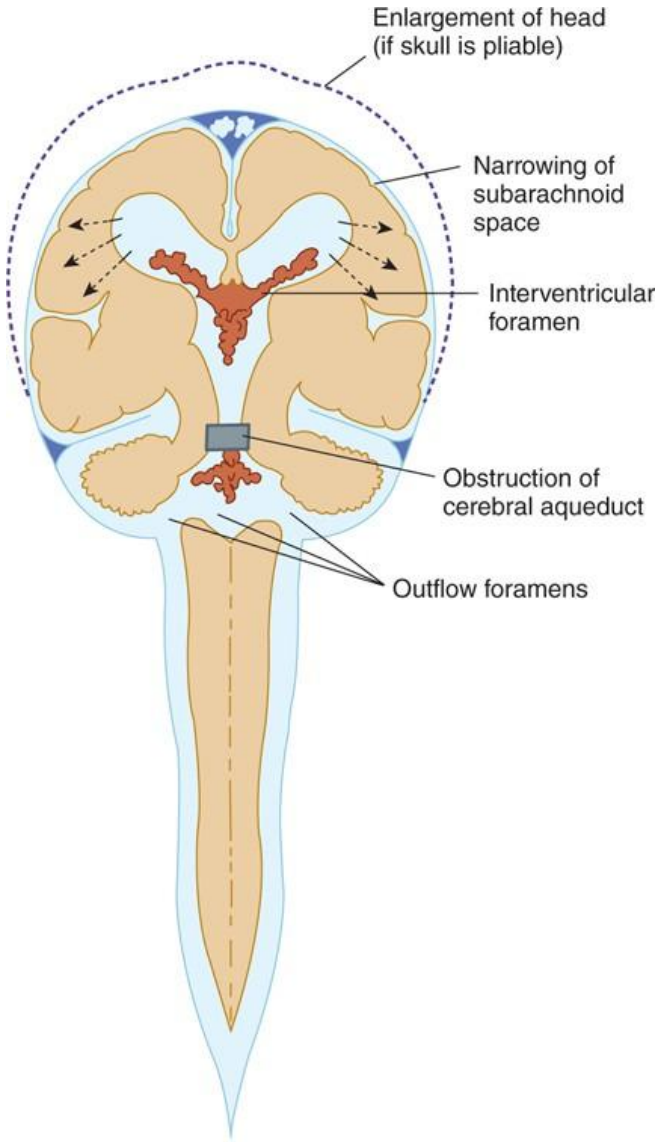
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Mechanical Obstruction

Aqueductal Stenosis, Chiari, Dandy-Walker

Aqueductal Stenosis

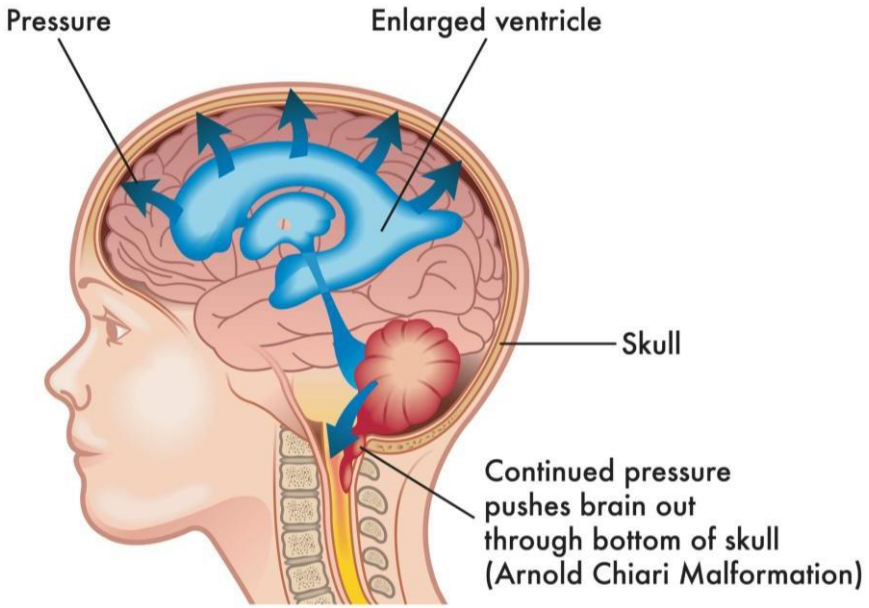
- Obstruction of the cerebral aqueduct
- Blocks CSF flow: 3rd → 4th ventricle
- Enlarged lateral and 3rd ventricles
- Normal 4th ventricle
- Congenital or acquired
- Infants: ↑ head circumference, signs of ↑ ICP
- Adults: ↑ ICP, headaches, ↓ consciousness



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Chiari II Malformation (Arnold-Chiari Malformation)

- Small/crowded posterior fossa ➔ hindbrain displaced downward
- Crowding impairs CSF flow ➔ noncommunicating (obstructive) hydrocephalus
- Strongly associated with *myelomeningocele*
- Infants may have feeding difficulty, weak cry, and signs of ↑ ICP



Chiari I

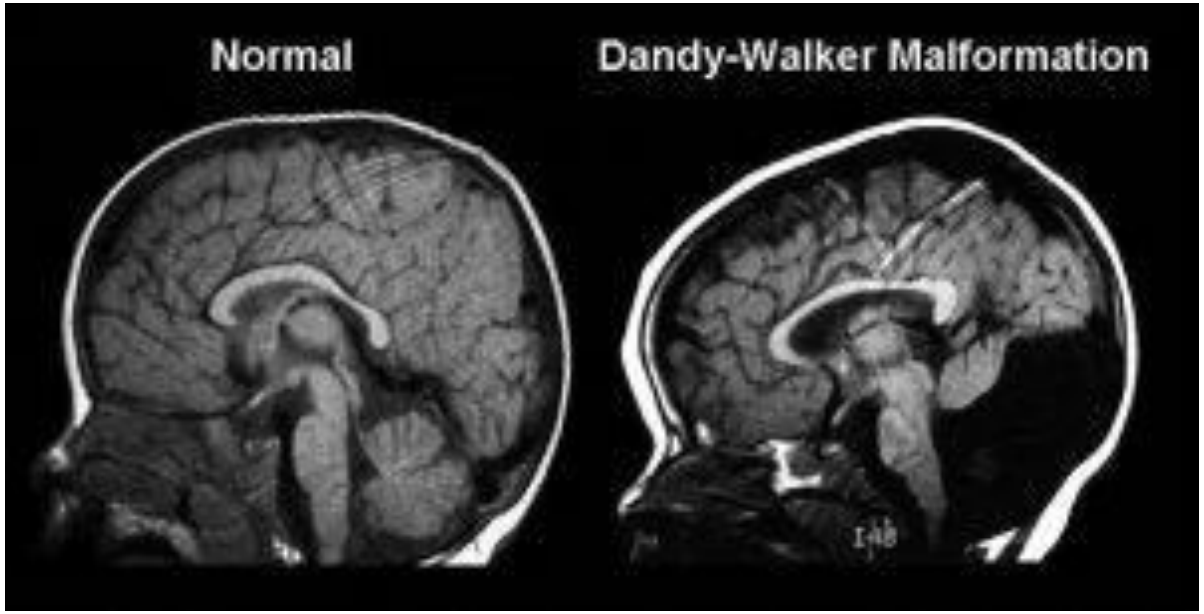
- Cerebellar tonsillar herniation
- Typically presents later
- No myelomeningocele

Chiari II = small posterior fossa + hindbrain herniation + myelomeningocele + hydrocephalus

Chiari I = tonsillar herniation only, no myelomeningocele

Dandy-Walker Malformation (DWM)

- Hypoplasia / agenesis of cerebellar vermis
- ➔ Cystically dilated 4th ventricle; enlarged posterior fossa
- ➔ impaired CSF flow
- ➔ Obstructive hydrocephalus



Clinical Features

- Hydrocephalus (macrocephaly)
- Delayed motor development and ataxia

Dandy-Walker = enlarged posterior fossa

Chiari II = crowded posterior fossa

Pressure Disorders

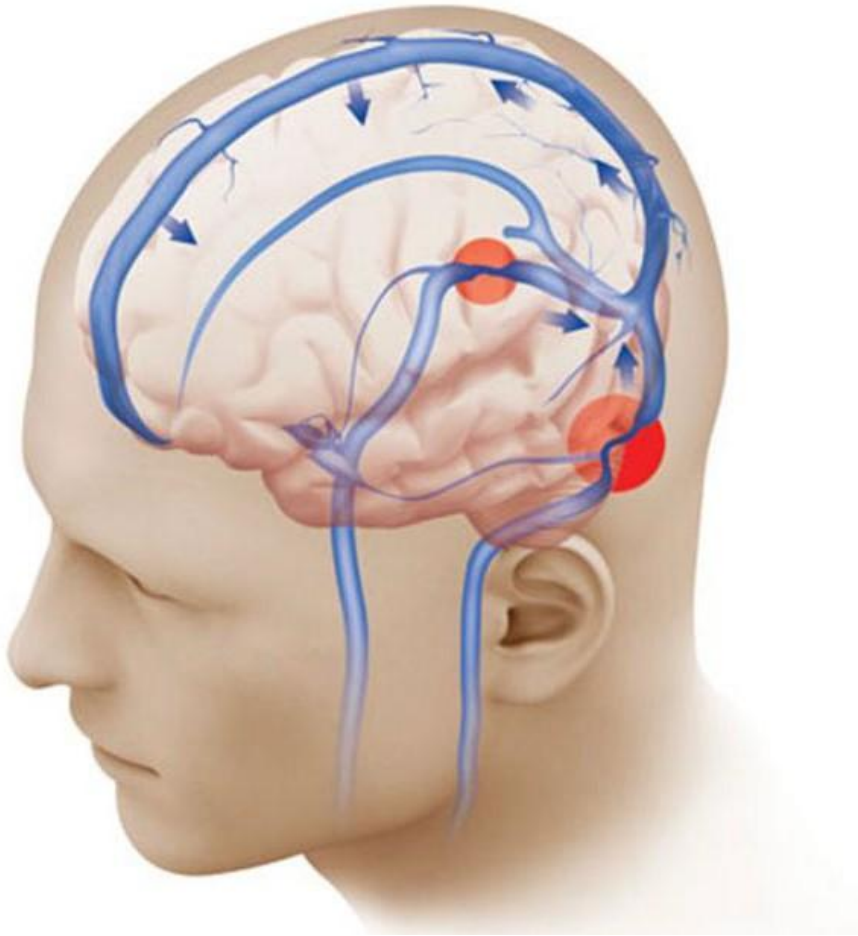
Idiopathic Intracranial Hypertension (pseudotumor Cerebri)

Normal Pressure Hydrocephalus (NPH)

Hydrocephalus Ex vacuo (ventriculomegaly due to brain atrophy; not true hydrocephalus)

Idiopathic Intracranial Hypertension (Pseudotumor Cerebri)

- ↑ ICP with normal ventricular size on imaging
- No mass lesion or hydrocephalus
- Classically occurs in young woman with obesity
- Headache, papilledema, transient visual loss
- Pulsatile tinnitus (key diagnostic clue)
- CN VI palsy may cause diplopia
- LP: ↑ opening pressure with normal CSF composition
- Tx: weight loss + acetazolamide
- Often associated with transverse venous narrowing



Some cases of pseudotumor cerebri may result from narrowing (red) of cerebral draining veins (blue).
Image: neurosurgery.weillcornell.org

Normal Pressure Hydrocephalus (NPH)

- Usually occurs in older adults
- Ventriculomegaly out of proportion to cortical atrophy

Classic Triad

- Urinary incontinence; gait disturbance; dementia

- CSF opening pressure may be normal when measured
- Tx: ventriculoperitoneal shunt

Traditional Explanation

- Thought to involve impaired CSF absorption

  ➔ ventricular enlargement compresses corona radiata

NPH = “wet, wobbly, wacky”
Gait disturbance typically occurs first



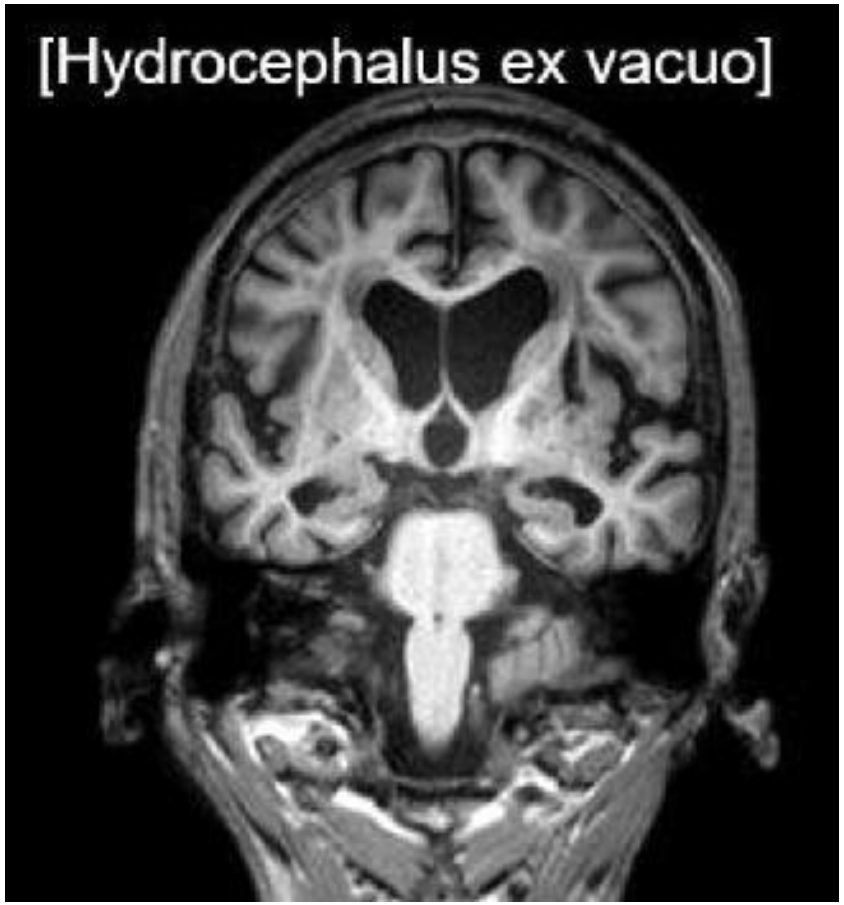
Source: Usatine RP, Smith MA, Mayeaux EJ, Chumley HS: *The Color Atlas of Family Medicine, Second Edition*: www.accessmedicine.com
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Non-contrast CT demonstrates enlarged lateral ventricles (*white arrows*) without significant cerebral atrophy. (*Courtesy of Reginald Dusing, MD.*)

Hydrocephalus Ex Vacuo

- Ventricular enlargement due to loss of brain tissue
- Compensatory enlargement of ventricles and sulci
- Not due to CSF obstruction or impaired absorption
- ICP is normal
- Seen in neurodegeneration (Alzheimer), stroke, or other causes of brain atrophy
- Despite the name, not true hydrocephalus

NPH: ventriculomegaly + clinical triad
Ex vacuo: ventriculomegaly + brain atrophy



Kim. *Psychiatry Investig*. 2021

Summary

- Meninges: dura → arachnoid → pia
- Spaces: epidural/subdural = potential; subarachnoid = CSF + vessels
- Hemorrhage: epidural = MMA; subdural = bridging veins; SAH = subarachnoid
- ↑ ICP → brain herniation across dural partitions
- CSF: choroid plexus → ventricles → subarachnoid space → venous sinuses
- Hydrocephalus: communicating = ↓ absorption; obstructive = blocked flow
- Key disorders have characteristic ventricular and imaging patterns

Meninges and Ventricular System

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